

Enhancing Security and Efficiency in Computational Hardware Through Intelligent Optimization



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Motivation

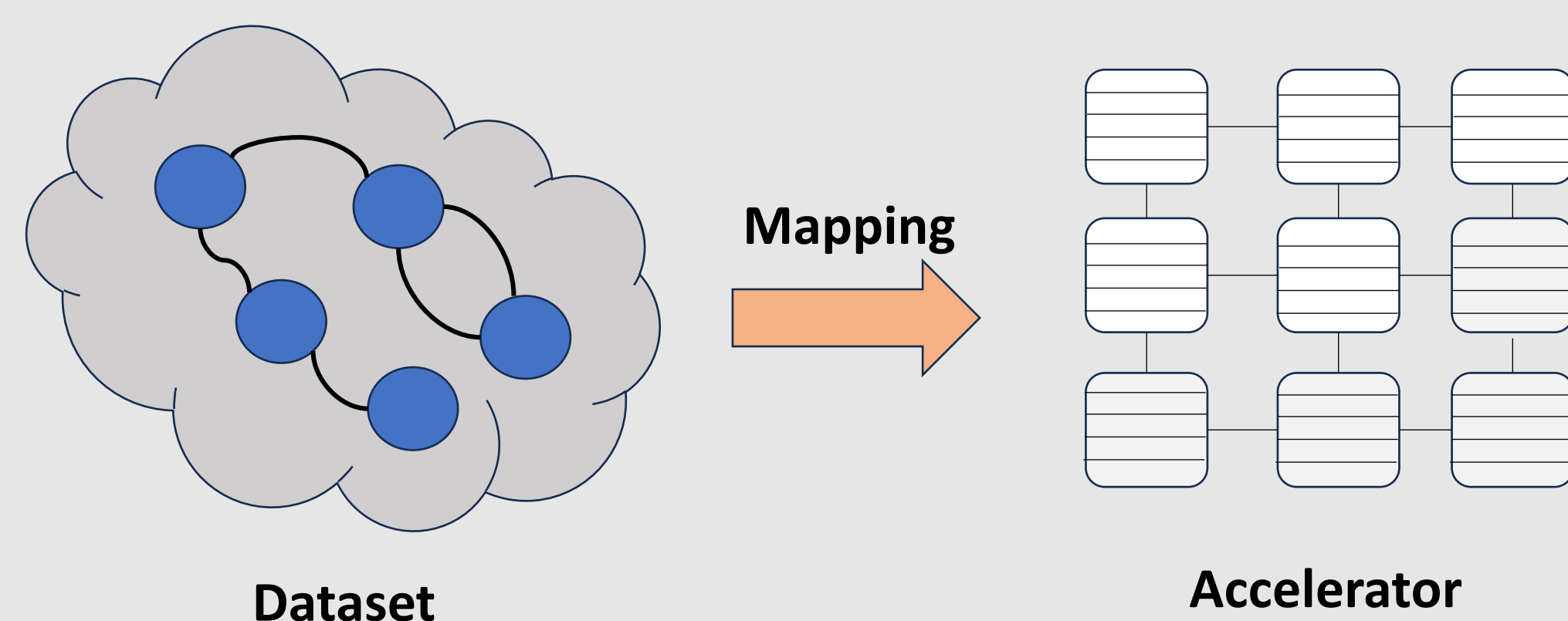
- Modern computational hardware, including CPUs, GPUs, and AI accelerators, relies on efficient memory access for performance and security.
- However, unnecessary memory accesses introduce vulnerabilities such as data leakage, side-channel attacks, and increased power consumption.
- Addressing these inefficiencies is crucial for ensuring both security and optimal resource utilization.

Problem

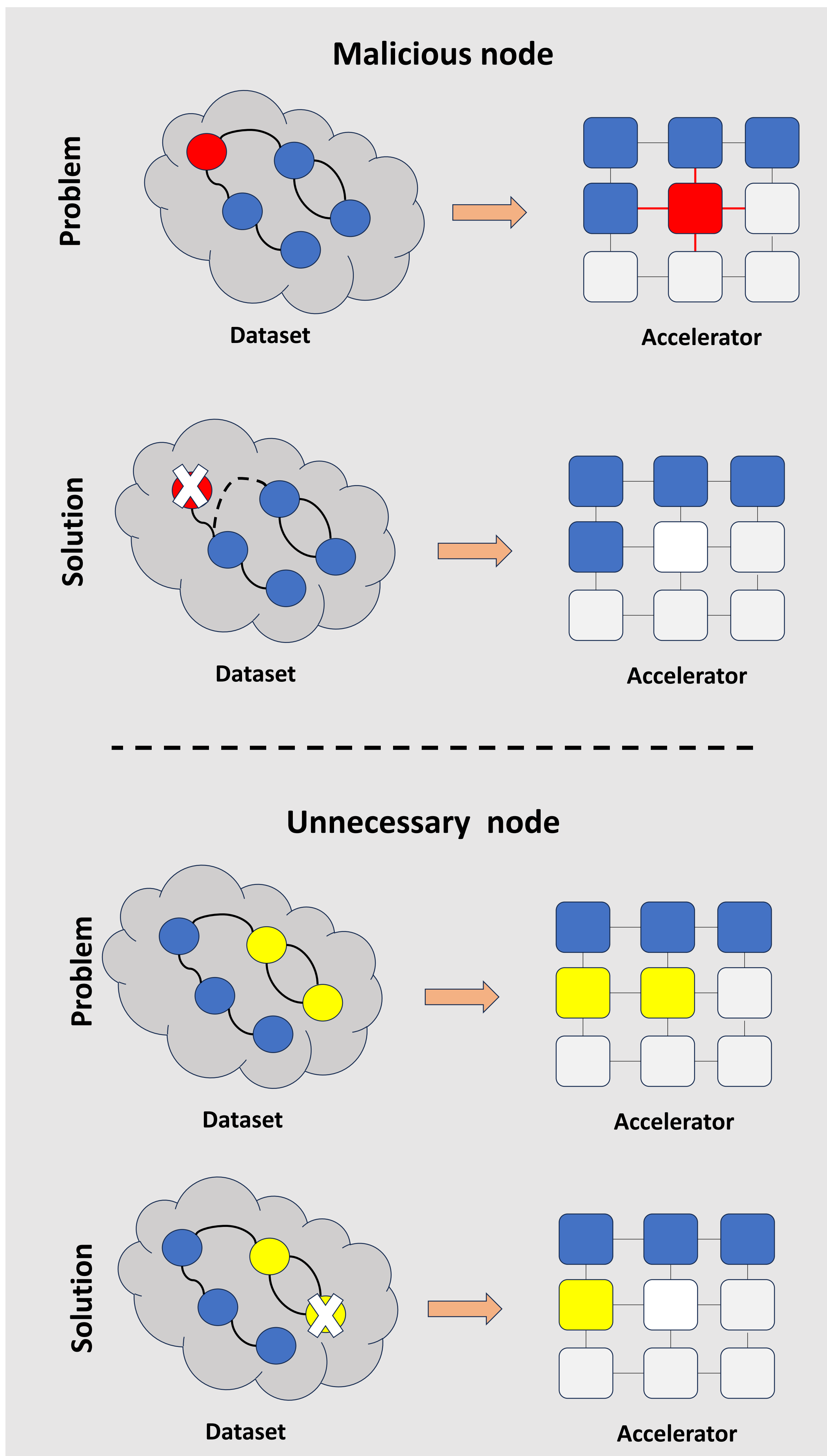
- Unnecessary or redundant memory accesses create security risks and degrade system efficiency.
- Data leakage:** Unused or poorly managed memory pathways provide opportunities for malicious code insertion and unauthorized data extraction
- Performance and Energy Overhead:** Frequent redundant memory accesses lead to higher latency, increased power consumption, and inefficient memory utilization, impacting computational workloads

Example

- NAPOLY is a two-dimensional array of processor elements where each processor element consists of RAM and gates.
- Datasets are mapped on the accelerator before starting the operation.
- Dataset represents labels/scores for each edge



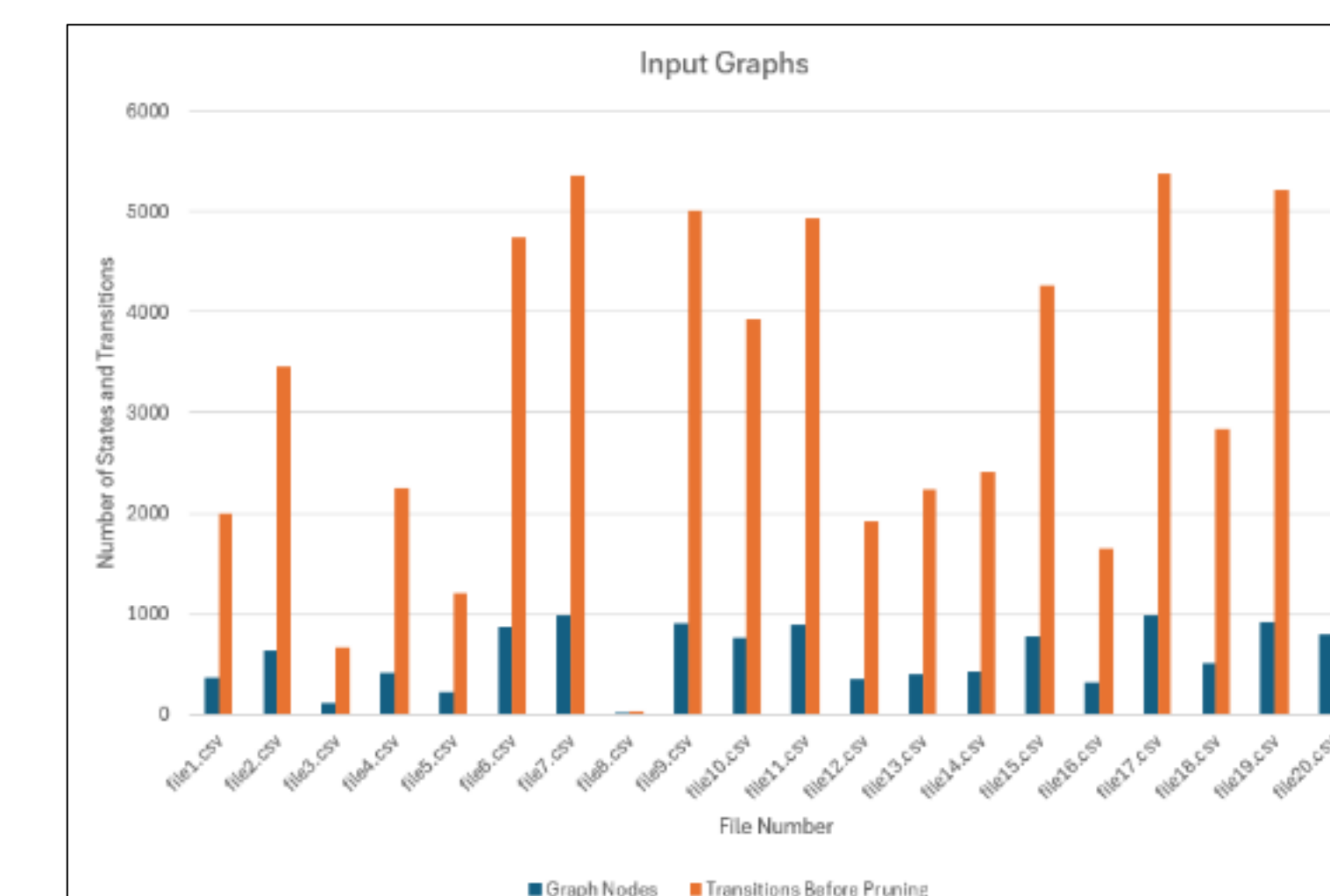
Explanation



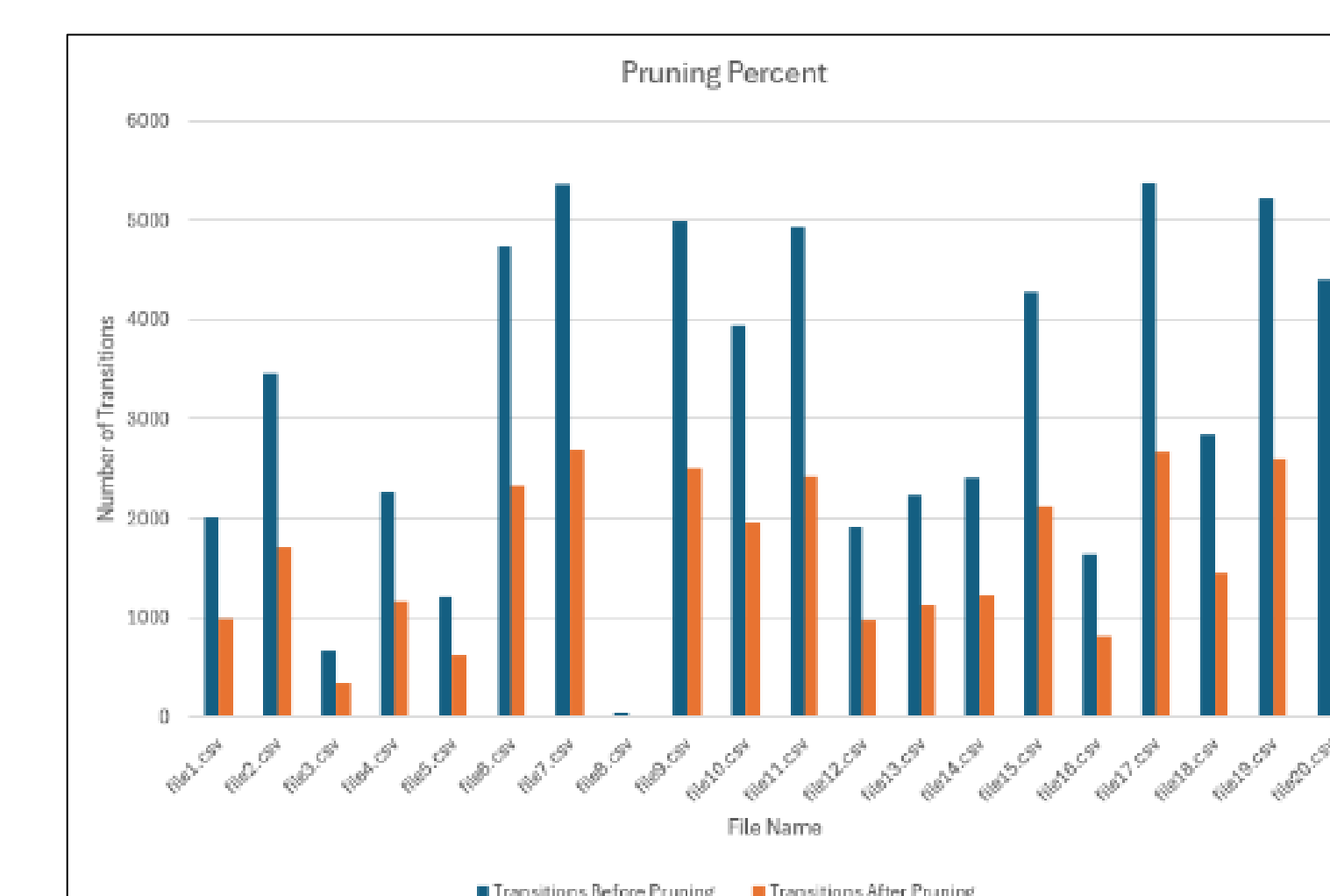
Proposed Work

- Develop an intelligent optimization method that eliminates unnecessary states/edges and deduct any malicious information before mapped on processor elements.
- Test and compare the results with state-of-the-art designs.

Preliminary Results



Dataset: 20 files
with #states =
10-10000



Results after
eliminating
weak nodes

Resources

- Karbowiczak, Ryan, and Rasha Karakchi. "Optimizing Sequence Alignment with Scored NFAs." *arXiv preprint arXiv:2501.02162* (2025). Run on existing tool to eliminate the redundancy
- Aramuthakannan, S., et al. "Secured Hardware Platform for FPGA AI Model Protection."

Acknowledgement

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